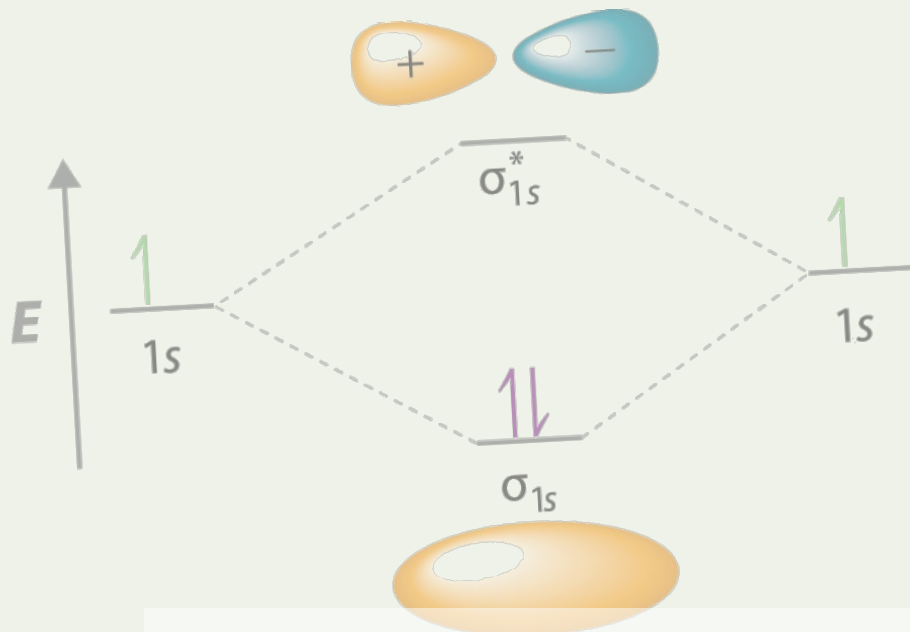




# Condensed Matter Theory



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# CHAPTER 1 Preparation

## 1.1 Outline

**Free electron gas** Drude model; Sommerfeld model; magneto-transport

**Electrons in a crystal** Crystal structure (such as symmetry); Diffraction experiments; Lattice vibrations

**Band theory** Bloch's theorem; Nearly free electron model (simplified model); Tight binding Model (simplified model); Applications of band theory; Semiconductor devices

**Exotic band theory** Dirac band; [Topological band; Flat band]

**Beyond band theory (Many-body physics)** Magnetism; Superconductivity

## 1.2 Course Overview

**Lecture 1** Introduction

**Lecture 2** Free electron model of metals

**Lecture 3** Lattice vibrations and phonons

**Lecture 4** Crystal symmetry and crystal structure

**Lecture 5** Scattering experiments (X-ray diffraction and electron diffraction)

**Lecture 6** Electron waves in crystals (Bloch's theorem)

*The two models help us to Understand why there's a gap in some crystal*

**Lecture 7** Band theory I: Nearly free electron model

**Lecture 8** Band theory II: Tight binding model

**Lecture 9** Band theory III: Fermi surface and effective mass

**Lecture 10** Physics of semiconductors

**Lecture 11** Electrons in a magnetic field: Quantum Hall effect

**Lecture 12** Dirac band theory I: band structure of graphene

**Lecture 13** Dirac band theory II: Landau quantization of graphene

**Lecture 14** Magnetism

**Lecture 15** Superconductivity

**Lecture 16** Topological Physics

### 1.3 References

- Introduction to Solid State Physics, Charles Kittel
- Solid State Physics, Neil Ashcroft and David Mermin
- The Oxford Solid State Basics, Steven H. Simon
- Solid State Physics: Essential Concepts, David W. Snoke

## CHAPTER 2

## CHAPTER 3 Free electron model of metals

### 3.1 Drude model (1900) for electrons in metals

### 3.2 Ohm's Law in macroscopic

$$V = \int_a^b E \, dx = Ed, \quad V = IR \quad (3.1)$$

Define current density  $j = I/A$ , therefore

$$Ed = jAR, \quad E = \left( \frac{RA}{d} \right) j$$

then, we can define the conductivity  $\sigma = 1/\rho$ , where  $\rho = RA/d$  is the resistivity. So,

$$\mathbf{j} = \sigma \mathbf{E} \quad (3.2)$$

### 3.3 Resistivity

1. Metals:
  - Very conductive
  - The resistivity increases with increasing temperature
2. Semiconductors
  - Limited conductance
  - The resistivity decreases with increasing temperature
3. Insulators: Very weak conductance

### 3.4 Sommerfeld model (1920s) for electrons in metals

### 3.5 Drude model

Without electric field:

- Free electron moving randomly
- $\frac{1}{2}mv_{\text{rms}}^2 = \frac{3}{2}k_B T$ ,  $v_{\text{rms}} = 1.2 \times 10^5 \text{ m/s}$  at room temperature.



With electric field, the drift velocity  $v_d \ll v_{\text{rms}}$ ,  $v_d \sim 10^{-5}$  m/s. The definition of the drift velocity is  $v_d = l/\tau$ .

Let  $\tau$  be the scattering time and  $1/\tau$  be the scattering rate. Consider in a time interval  $dt$ , a fraction  $dt/\tau$  of the electrons are scattered, and their average velocity becomes zero.

### 3.6 Classical Hall effect

### 3.7 Drude model with a magnetic field

Consider Drude model with applied electric and magnetic fields:

$$m^* \dot{\mathbf{v}} = -e(\mathbf{E} + \mathbf{v} \times \mathbf{B}) - m^* \mathbf{v}/\tau$$

In steady state, we have

$$\mathbf{E} = -\frac{m^* \mathbf{v}}{e\tau} - \mathbf{v} \times \mathbf{B}$$

and we can rewrite it into 3 components:

$$\begin{aligned} \frac{v_x}{\tau} - \omega_c v_y &= -\frac{e}{m} E_x \\ \frac{v_y}{\tau} + \omega_c v_x &= -\frac{e}{m} E_y \\ \frac{v_z}{\tau} &= -\frac{e}{m} E_z \end{aligned}$$

then we arrive at

$$\begin{aligned} v_x &= -\frac{e\tau}{m(1 + \omega_c^2 \tau^2)} (E_x + \omega_c \tau E_y) \\ v_y &= -\frac{e\tau}{m(1 + \omega_c^2 \tau^2)} (E_y - \omega_c \tau E_x) \\ v_z &= -\frac{e\tau}{m} E_z \end{aligned}$$

and we can write it into matrix form

$$\begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} = \hat{\sigma} \begin{pmatrix} E_x \\ E_y \\ E_z \end{pmatrix}$$

or  $\hat{\rho} = \sigma^{-1}$ .

### 3.8 Sommerfeld Model

#### 3.8.1 Electron wave

Time-independent Schrödinger equation

$$-\frac{\hbar^2}{2m} \nabla^2 \psi(\mathbf{r}) + U\psi(\mathbf{r}) = E\psi(\mathbf{r}) \quad (3.3)$$

For a free electron:  $U = 0$ ,

$$\nabla^2 \psi(\mathbf{r}) = -\frac{2mE}{\hbar^2} \psi(\mathbf{r})$$

The solution is a plane wave

$$\psi(\mathbf{r}) = \frac{1}{\sqrt{V}} e^{i(k_x x + k_y y + k_z z)} \quad (3.4)$$

Energy

$$E = \frac{\hbar^2 k^2}{2m} = \frac{\hbar^2 (k_x^2 + k_y^2 + k_z^2)}{2m} \quad (3.5)$$

apply the Born-von Karman periodic boundary conditions

$$\begin{aligned} \psi(x + L_x, y, z) &= \psi(x, y, z) \\ \psi(x, y + L_y, z) &= \psi(x, y, z) \\ \psi(x, y, z + L_z) &= \psi(x, y, z) \end{aligned} \quad (3.6)$$

States in  $k$ -space

$$k_x = \frac{2\pi n_1}{L_x}, \quad k_y = \frac{2\pi n_2}{L_y}, \quad k_z = \frac{2\pi n_3}{L_z} \quad (3.7)$$

with  $n_i \in \mathbb{Z}$ . In each unit (3D grid) with size  $d^3$ :  $\Delta d = \frac{2\pi}{L_x}[(x+1) - (-x)]$ . The volume is  $\frac{2\pi}{L_x} \cdot \frac{2\pi}{L_y} \cdot \frac{2\pi}{L_z}$ , there are  $2 \times V/(2\pi)^3$  quantum states per unit volume in  $k$ -space, while the density is  $\frac{2}{V} = 2 \times \frac{1}{\left(\prod_i \frac{2\pi}{L_i}\right)}$ .

One can simply think that the allowed states inside a sphere. The volume of the sphere should be

$$V = \frac{4}{3}\pi k_F^3$$

while the density is

$$2 \times \frac{1}{\left(\frac{2\pi}{L}\right)^3}$$

then, the number of electrons in the sphere should be

$$N = 2 \times \frac{(4\pi/3)k_F^3}{(2\pi/L)^3} = \frac{k_F^3}{3\pi^2} V$$

We define the *Fermi wave vector*:  $k_F = (3\pi^2 n)^{1/3}$ , which we can compute it from the carrier density  $n = N/V$ .

### 3.9 Bosons and Fermions at $T \neq 0$

Bosons:

$$\bar{n}_i = \frac{g_i}{\exp} \quad (3.8)$$

at  $T = 0$ ,

$$f(E) = \begin{cases} 0, & \text{for } E > E_f \\ 1, & \text{for } E < E_f \end{cases}$$

### 3.10 Fermi-Dirac Distribution

$$f(E) = \frac{1}{1 + \exp[(E - E_f)/k_B T]} \quad (3.9)$$

### 3.11 Density of states

$$n(T) = \int f(E)g(E) dE$$

$g(E)$ : how many states are allowed at energy  $E$ .

$$g(E) = \frac{1}{V} \frac{dN}{dE} \quad (3.10)$$

we calculate the total number

$$N = \frac{k^3}{3\pi^2} V \quad (3.11)$$

and

$$E = \frac{\pi^2 k^2}{2n} \quad (3.12)$$

Substitute  $E$  and  $N$  into  $g(E) = \frac{1}{V} \frac{dN}{dE}$ , we can have

$$g(E) = \frac{mk}{\hbar^2 \pi^2} \quad (3.13)$$

which  $k = \frac{\sqrt{2mE}}{\hbar}$ .

$$(a) \quad g_{3D}(E) = \frac{m^*}{\pi^2 \hbar^3} \sqrt{2m^* E} \propto E^{1/2}$$

$$(c) \quad g_{1D}(E) = \frac{1}{\pi \hbar} \sqrt{\frac{2m^*}{E}} \propto E^{-1/2}$$

$$(b) \quad g_{2D}(E) = \frac{m^*}{\pi \hbar^2} = \text{Const}$$

$$(d) \quad g_{0D}(E) = 2\delta(E)$$

The total number of electron at  $T > 0$  per volume

$$\frac{N}{V} = \int \underbrace{f(E)}_{\text{probability}} \cdot \underbrace{g(E) dE}_{\text{can be occupied}}$$

and the energy for the electrons at the Fermi energy level (internal energy) is

$$\bar{E} = \int E f(E) g(E) dE \quad (3.14)$$

to calculate the velocity

$$v(E) = \int v(E) f(E) g(E) dE, \quad \text{and} \quad v(k) = \int v(k) f(k) g(k) dk$$

remember to add a  $\frac{V}{(2\pi)^3}$  factor when calculate in  $k$ -space. Then, the current density is

$$j = -nev = -2e \int \frac{d^3 \mathbf{k}}{(2\pi)^3} f(\mathbf{k}) v(\mathbf{k}) \quad (3.15)$$

Consider  $\mathbf{j} = \sigma \mathbf{E}$ , when  $\mathbf{E} = \mathbf{0}$  then  $\mathbf{j} = \mathbf{0}$ . Now, check it. At zero field, for every occupied state  $\mathbf{k}$ , there is a state  $-\mathbf{k}$  occupied with the same probability and the corresponding velocity is same but negative. For the current density in an applied electric field,

$$\mathbf{j} = -nev = -2e \int \frac{d^3\mathbf{k}}{(2\pi)^3} f\left(\mathbf{k} + \frac{e\tau}{\hbar} \mathbf{E}\right) v(\mathbf{k}) = -2e \int \frac{d^3\mathbf{k}}{(2\pi)^3} f(\mathbf{k}) v\left(\mathbf{k} - \frac{e\tau}{\hbar} \mathbf{E}\right) \quad (3.16)$$

since  $v(\mathbf{k}) = \hbar\mathbf{k}/m$

$$\mathbf{j} = \frac{e^2\tau}{m} \left[ 2 \int \frac{d^3\mathbf{k}}{(2\pi)^3} f(\mathbf{k}) \right] \mathbf{E} = \frac{e^2\tau}{m} n \mathbf{E} = \sigma \mathbf{E}$$

which  $\sigma = ne\tau^2/m$  is the same as that in Drude mode.

### 3.11.1 Heat capacity of an electron gas

The change of internal energy when heated from  $0K$  to  $T > 0$

$$\Delta U = U(T) - U(0) = \int - \int \quad (3.17)$$

- At Mediate temperature:  $T^3$
- At Low temperature:  $T$
- At High temperature: Const

At low temperature,  $f$  choose to be step function, and obviously,  $\frac{\partial}{\partial T} f$  will be like a  $\delta$ -function.

### 3.11.2 Thermal transport

## 3.12 Limitation of Free electron gas model

## CHAPTER 4 Lattice Vibrations and Phonons

### 4.1 Specific heat capacity in solids

$$m \frac{d^2 u_n}{dt^2} = \beta(u_{n+1} + u_{n-1} - 2u_n) \quad (4.1)$$

should have wave-like solutions

$$u_n = A e^{i(\omega t - nak)}$$

Substitute it to the ODE, then we have the dispersion relation

$$\omega = 2\sqrt{\frac{\beta}{m}} \left| \sin\left(\frac{ka}{2}\right) \right|$$

- The dispersion relation is periodic in  $k$ , with period  $2\pi/a$
- ...

$k = n \frac{2\pi}{Na}$  For a  $n$ -number of particles, we have  $n$  phonons, each phonon has a vibration, that is a  $\omega$ .

For small  $ka$ ,  $\omega \approx a\sqrt{\frac{\beta}{m}}|k|$ .

$$k \in [-\pi/a, \pi/a], k \rightarrow k + 2\pi/a$$

$$\hbar \frac{\pi}{a} \times 3 \rightarrow \frac{2\pi}{a} \hbar$$

$$-\frac{2}{3} \hbar \frac{\pi}{a} \times 3 - \frac{2\pi}{a} \hbar + \frac{2\pi}{a} \hbar \times 2$$

#### 4.1.1 A diatomic chain

We have two kind of atoms, and keep the period. Using two kind of equations

$$m \frac{d^2 u_{2n}}{dt^2} = \beta(u_{2n+1} + u_{2n-1} - 2u_{2n}) \quad (4.2)$$

$$M \frac{d^2 u_{2n+1}}{dt^2} = \beta(u_{2n+2} + u_{2n} - 2u_{2n+1}) \quad (4.3)$$

Similarly

$$u_{2n} = A e^{i(\omega t - 2nak)}$$

#### 4.1.2 Gap

For gap: at  $k = \frac{\pi}{2a}$ , supposed  $M > m$ , we have

$$\omega_+^2 = \frac{2\beta}{m} > \omega_-^2 = \frac{2\beta}{M}$$

when  $m = M$ , the gap equals to zero.

# CHAPTER 5 Crystal Symmetry & Crystal Structure

## 5.1 Crystal Symmetry

### 5.1.1 Inversion Symmetry

### 5.1.2 Rotoinversion Symmetry

## 5.2 Crystal Structure

$$\text{Crystal lattice} + \text{Basis} = \text{Crystal structure} \quad (5.1)$$

Replace a every site (represent the symmetry of crystals) by basis (represent the compound of crystal: one / a group of atom / a molecule / a set of molecule): nvotif

Fourier transfor:

$$e^{i\mathbf{G} \cdot \mathbf{R}} = 1, \quad A(\mathbf{k}) = A(\mathbf{k}) e^{i\mathbf{k} \cdot \mathbf{R}} \quad (5.2)$$

then  $e^{i\mathbf{k} \cdot \mathbf{R}} = 1$ .

$$\mathbf{a}_i \cdot \mathbf{b}_j = 2\pi \delta_{ij} \quad (5.3)$$

**Example 5.2.1** (Rec. lattice of BCC).

### 5.2.1 Structure factor of a monatomic lattice

In BCC structure, there are two atoms in a unit cell:  $(0, 0, 0)$  and  $\frac{a}{2}(1, 1, 1)$ . They have the same atomic form factor  $f$

$$S_{\mathbf{G}} = f e^{-i\mathbf{G}_1 \cdot \mathbf{r}_1} + f e^{-i\mathbf{G}_2 \cdot \mathbf{r}_2}$$

where  $\mathbf{G}_1 = \frac{2\pi}{a}(hx_0 + ky_0 + lz_0)$  and  $\mathbf{G}_1 \cdot \mathbf{r}_1 = 0$ ;  $\mathbf{G}_2 = \frac{2\pi}{a}(1 \times \frac{a}{2}h + 1 \times \frac{a}{2}k + 1 \times \frac{a}{2}l)$ .

- $S_{\mathbf{G}} = 0$ , when  $h + k + l$  is an odd integer, there No diffraction.
- $S_{\mathbf{G}} = 2f$ , when  $h + k + l$  is an even integer, diffraction peak exists.

X-Ray Diffraction: some peaks in  $I$  along  $\theta$ : stands for  $(100)$ ,  $(111)$ ,  $(110)$ , but  $(110)$  is missing. *Missing orders make it easier to identify a particular crystal structure.*

The wave length of the X-Ray source has a range  $(\lambda_1, \lambda_2)$ , corresponding  $k_{\max}$  to  $k_{\min}$ . For each  $k$ , we have a circle 'field'.

For incident beam, we can get a series of ring: the distance between the rings  $d = a/\sqrt{h^2 + k^2 + l^2}$  for cubic lattice.

### 5.2.2 Bloch theorem

Lattice vector

$$\mathbf{R} = k_1 \mathbf{x}_1 + k_2 \mathbf{x}_2 + k_3 \mathbf{x}_3$$

where  $k_i$  should be integer numbers. Bloch electrons

$$\left[ -\frac{\hbar^2}{2m} \nabla^2 + U(\mathbf{r}) \right] \psi(\mathbf{r}) = E \psi(\mathbf{r})$$

where  $U(\mathbf{r} + \mathbf{R}) = U(\mathbf{r})$ . The solution should be the form of

$$\psi_k(\mathbf{r}) = e^{i\mathbf{k} \cdot \mathbf{r}} u_k(\mathbf{r}), \quad (5.4)$$

where  $u_k(\mathbf{r}) = u_k(\mathbf{r} + \mathbf{R})$ , multiplication of Plane wave and lattice periodic function.

### 5.2.3 Nearly free electron model

For free electrons, the periodic function

$$E = \frac{\hbar^2 k^2}{2m} \quad (5.5)$$

then, when we apply the Bloch theorem

$$\psi_R(\mathbf{r}) = e^{i\mathbf{k} \cdot \mathbf{r}} u_k(\mathbf{r}), \quad u_k(\mathbf{r}) = u_k(\mathbf{r} + \mathbf{R})$$

**Example 5.2.2** (Simple cube).

$$E(\mathbf{k}) = \epsilon_s - J_0 - 2J_1(\cos(k_x a) + \cos(k_y a) + \cos(k_z a))$$

where

$$-\mathbf{k} \cdot \mathbf{R} = -k_x \cdot a + k_y \cdot 0 + k_z \cdot 0 = -k_x \cdot a$$

In the first Brillouin zone of simple cube, there are several high symmetric points

- |                                               |                                                      |
|-----------------------------------------------|------------------------------------------------------|
| (a) $\Gamma$ points: $\mathbf{k} = (0, 0, 0)$ | (c) $R$ points: $\mathbf{k} = (\pi/a, \pi/a, \pi/a)$ |
| (b) $X$ points: $\mathbf{k} = (\pi/a, 0, 0)$  | (d) $M$ points: $\mathbf{k} = (\pi/a, \pi/a, 0)$     |

Along different directions, the band structure is different.

$$E(\Gamma) = \epsilon_0 - J_0 - 6J_1, \quad (5.6)$$

$$E(X) = \epsilon_0 - J_0 - 2J_1, \quad (5.7)$$

$$E(R) = \epsilon_0 - J_0 + 6J_1, \quad (5.8)$$

The effective mass. Around the “minimum” point, the electrons behave as the free electrons, i.e.,

$$E = \frac{\hbar^2 k^2}{2m}$$

Since we have (for cubic)

$$E(k) = \epsilon_0 - J_0 - 6J_1 + J_1|\mathbf{k}|^2 a^2$$

and we can expand it into Taylor series

$$E = E_0 + a_1 \frac{\partial E}{\partial k} + a_2 \frac{\partial^2 E}{\partial k^2} + \dots$$

where  $\frac{\partial E}{\partial k}$  around the minimum point. Compare the expressions, we have the effective mass

$$m^* = \frac{\hbar^2}{\partial^2 E / \partial k^2}$$

The bands have antisymmetric properties, i.e.,

$$\frac{\partial E}{\partial k_x} \neq \frac{\partial E}{\partial k_y} \neq \frac{\partial E}{\partial k_z}, \quad \text{and} \quad \frac{\partial^2 E}{\partial k_x^2} \neq \frac{\partial^2 E}{\partial k_y^2} \neq \frac{\partial^2 E}{\partial k_z^2},$$

## 5.3 Bloch Oscillations

For three dimensional

$$E = \epsilon_0 - J_0 - 2J_1(\cos(k_x a) + \cos(k_y a) + \cos(k_z a))$$

For one dimensional

$$E = \epsilon_0 - J_0 - 2J_1 \cos(k_x a)$$

### 5.3.1 Requirements for observation

$$w = \frac{eEa}{\hbar}, \quad T = \frac{2\pi}{eEa} = \frac{h}{eEa}$$



## CHAPTER 6

## CHAPTER 7

## CHAPTER 8 Band theory II: Tight binding model

### 8.1 Metal, insulator

### 8.2 Excitation

Starting from the Schrödinger equation including electron interactions

$$\left[ -\frac{\hbar^2}{2m_e^*} \nabla_e^2 - \frac{\hbar^2}{2m_h^*} \nabla_h^2 - \frac{e^2}{4\pi\epsilon\epsilon_0|r_e - r_h|} \right] \psi_{ex} = E\psi_{ex} \quad (8.1)$$

To solve it, extract the eigenvalue  $E$ . The Hamiltonian can be separated into two parts: the free electron part and the Hydrogen part

$$E_n^{\text{ex}} = E_g - \frac{m_0^* e^4}{2(4\pi\epsilon\epsilon_0)^2 \hbar^2 n^2} = E_g - \frac{R_{\text{ex}}}{n^2} \quad (8.2)$$

where  $R_{\text{ex}}$  denotes the excitation Rydberg.

## CHAPTER 9 Physics of semiconductors

### 9.1 Intrinsic semiconductors

The number of particles

$$n = \int_{E_c}^{\infty} dE g(E) f(E) \quad (9.1)$$

where

$$g(E) = \frac{m_n^*}{\pi^2 \hbar^3} \sqrt{2m_n^*(E - E_c)}$$

Typically,  $E_f$  is inside the band gap. In most cases, the energy  $E \gg E_f$ . So, the distribution function is

$$f(E) = \frac{1}{e^{(E-E_f)/kT} + 1} \approx e^{-(E-E_f)/kT}$$

Then, the integral

$$n = \int_{E_c}^{\infty} \frac{m^*}{\pi^2 \hbar^3} \sqrt{2m^*(E - E_c)} e^{-(E-E_f)/kT} dE = N_c e^{-(E_c-E_f)/kT} \quad (9.2)$$

where  $N_c = \frac{2(2\pi m_c^* kT)^{3/2}}{\hbar^3}$ .

One can simplify understanding

$$n_i^2 = n \cdot p = N_c \cdot N_v e^{-E_g/kT} \quad (9.3)$$

$n$  and  $p$  satisfies

$$\begin{cases} n + N_a &= p + N_d, \\ np &= n_i^2 \end{cases}$$

then, we can solve

$$n = \frac{N_d - N_a}{2} + \left[ \left( \frac{N_d - N_a}{2} \right)^2 + n_i^2 \right]^{1/2}, \quad (9.4)$$

$$p = \frac{N_a - N_d}{2} + \left[ \left( \frac{N_a - N_d}{2} \right)^2 + n_i^2 \right]^{1/2}, \quad (9.5)$$

Case 1  $n$ -type:  $N_d - N_a \gg n_i$

Case 2  $p$ -type:  $N_d - N_a \ll n_i$

## Lecture #1 Free electron model

**Problem 1.1.** Two-charge-carrier Drude model:

Consider a system of two types of charge carriers in the Drude model. The two carriers have the same density ( $n$ ) and opposite charge ( $e$  and  $-e$ ), and their masses and relaxation rates are  $m_1, m_2$  and  $\tau_1, \tau_2$ , respectively. So their corresponding mobilities are  $\mu_1 = \frac{e\tau_1}{m_1}$  and  $\mu_2 = \frac{e\tau_2}{m_2}$ , respectively.

- Calculate the magnetoresistance  $\Delta\rho = \rho(B) - \rho(B=0)$ , where  $B$  is the magnetic field.
- Calculate the Hall coefficient.
- In an undoped semiconductor,  $n = n_0 e^{-\Delta/k_B T}$  describes the temperature dependence of the carrier concentration. What will the temperature dependence of the magnetoresistance and the Hall coefficient be?

**Solution.**

- For the two carriers, we have

$$\sigma_{0i} = \frac{ne^2\tau_i}{m_i} = ne\mu_i, \quad \omega_{ci} = \frac{eB}{m_i}, \quad \omega_{ci}\tau_i = \mu_i B$$

The total conductivity tensor is

$$\hat{\sigma} = \hat{\sigma}_1 + \hat{\sigma}_2$$

and the components are

$$\begin{aligned} \sigma_{xx} &= \frac{ne\mu_1}{1 + (\mu_1 B)^2} + \frac{ne\mu_2}{1 + (\mu_2 B)^2} \\ \sigma_{xy} &= \frac{ne\mu_1^2 B}{1 + (\mu_1 B)^2} + \frac{ne\mu_2^2 B}{1 + (\mu_2 B)^2} \\ \sigma_{zz} &= ne(\mu_1 + \mu_2) \end{aligned}$$

The resistance tensor is the inverse

$$\hat{\rho} = \hat{\sigma}^{-1} = \frac{1}{\sigma_{xx}^2 + \sigma_{xy}^2} \begin{pmatrix} \sigma_{xx} & -\sigma_{xy} & 0 \\ \sigma_{xy} & \sigma_{xx} & 0 \\ 0 & 0 & \frac{\sigma_{xx}^2 + \sigma_{xy}^2}{\sigma_{zz}} \end{pmatrix}.$$

Then, the element  $\rho_{xx}$  becomes

$$\rho_{xx}(0) = \sigma_{xx}^{-1}(0) = \frac{1}{ne(\mu_1 + \mu_2)}, \quad \text{and} \quad \rho_{xx} = \frac{\sigma_{xx}(B)}{\sigma_{xx}^2(B) + \sigma_{xy}^2(B)}$$

So, the magnetoresistance is

$$\Delta\rho = \rho_{xx}(B) - \rho_{xx}(0) = \frac{1}{ne} \left[ \frac{\frac{\mu_1}{1+(\mu_1 B)^2} + \frac{\mu_2}{1+(\mu_2 B)^2}}{\left( \frac{\mu_1}{1+(\mu_1 B)^2} + \frac{\mu_2}{1+(\mu_2 B)^2} \right)^2 + B^2 \left( \frac{\mu_1}{1+(\mu_1 B)^2} - \frac{\mu_2}{1+(\mu_2 B)^2} \right)^2} - \frac{1}{\mu_1 + \mu_2} \right]$$

(b) Since

$$\rho_{xy}(B) = -\frac{\sigma_{xy}(B)}{\sigma_{xx}(B)^2 + \sigma_{xy}(B)^2},$$

substitute it into the definition of Hall coefficient  $R_H = \frac{\rho_{xy}}{B}$ , that is

$$R_H = -\frac{1}{ne} \cdot \frac{\frac{\mu_1}{1+(\mu_1 B)^2} + \frac{\mu_2}{1+(\mu_2 B)^2}}{\left(\frac{\mu_1}{1+(\mu_1 B)^2} + \frac{\mu_2}{1+(\mu_2 B)^2}\right)^2 + B^2 \left(\frac{\mu_1}{1+(\mu_1 B)^2} - \frac{\mu_2}{1+(\mu_2 B)^2}\right)^2}$$

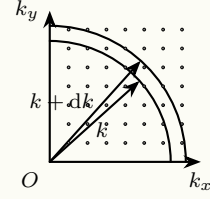
(c) For simplicity, consider the case  $B$  is small. Now,  $\Delta\rho$  and  $R_H$  becomes

$$\Delta\rho = ne\mu_1\mu_2(\mu_1 + \mu_2) \left(\frac{\mu_1 - \mu_2}{\mu_1 + \mu_2}\right)^2 B^2, \quad \text{and} \quad R_H = -\frac{1}{ne} \frac{\mu_1 - \mu_2}{\mu_1 + \mu_2}$$

So, in this case, the magnetoresistance  $\Delta\rho \propto e^{-\Delta/k_B T}$ , while the Hall coefficient  $R_H \propto e^{\Delta/k_B T}$ .

**Problem 1.2.** Density of States in 2D system:

Consider free electron in a 2D box with a width of  $a$ . The allowed wave-numbers  $k_x$  and  $k_y$  for  $x$  and  $y$  directions respectively are expressed as  $\frac{\pi n_x}{a}$  and  $\frac{\pi n_y}{a}$ , where  $n_x, n_y = 1, 2, 3, \dots$  are integer numbers. The following figure shows the schematic view of the 2D array of allowed quantum states in  $k$ -space.



- Determine the area in  $k$ -space occupied by one allowed  $k$  point.
- Express number of allowed states in the area enclosed by the first quarter circles with the radius of  $k$  and  $k + dk$ , where  $k \gg dk$  (Consider the spin degeneracy).
- Express density of allowed states in real space per unit energy.

**Solution.**

- The spaces of unit  $k$  point in  $k_x$ -direction and  $k_y$ -direction are

$$\Delta k_x = \frac{\pi}{a}, \quad \text{and} \quad \Delta k_y = \frac{\pi}{a}$$

Therefore, the area in  $k$ -space occupied by one allowed  $k$  point is

$$\Delta k_x \cdot \Delta k_y = \left(\frac{\pi}{a}\right)^2$$

- The area enclosed by the two quarter circles is

$$dA = \frac{1}{4} 2\pi k dk = \frac{\pi k}{2} dk$$

Since every  $k$  point occupy an area of  $(\pi/a)^2$ , so the number of allowed  $k$ -points in this area is

$$dN = \frac{2 dA}{(\pi/a)^2} = \frac{ka^2}{\pi} dk$$

Since two electrons can occupy each grid point ( $|\uparrow\rangle$  and  $|\downarrow\rangle$ ).

- Derivate it to the energy for a free election, we get

$$dE = d\left(\frac{\hbar^2 k^2}{2m}\right) = \frac{\hbar^2 k}{m} dk, \quad dk = \frac{m dE}{\hbar^2 k^2}.$$

Substitute it into the result in (b): the number of states in  $dk$

$$dN = \frac{ka^2}{\pi} dk = \frac{ma^2}{\pi \hbar^2} dE,$$

means in the whole system of area  $a^2$ , a number  $dN$  of states are allowed in energy range  $dE$ . So, the density of allowed states in real space per unit energy is

$$g(E) = \frac{1}{a^2} \frac{dN}{dE} = \frac{m}{\pi \hbar^2}.$$

**Problem 1.3.** Density of states from a general formula:

- (a) In a 3D solid and starting from the general formula,  $g(E) = \frac{2V}{(2\pi)^3} \iint \frac{dS_E}{\nabla_k E}$ , find the expression for the electronic density of states: (i) for free electrons; (ii) for electrons that obey the isotropic dispersion relation:  $E = -E_0 - 2\gamma \cos\left(\frac{ka}{2}\right)$ , where  $E_0, \gamma > 0$ .
- (b) After having established the corresponding expression for 2D, find  $g(E)$  for (i) and (ii) above.
- (c) Same question for 1D.

**Solution.** Firstly, we shall calculate something before: For the isotropic dispersion

$$E = -E_0 - 2\gamma \cos(ka/2),$$

we can obtain

$$\sin\left(\frac{ka}{2}\right) = \sqrt{1 - \left(\frac{E + E_0}{2\gamma}\right)^2}, \quad \text{and} \quad k = \frac{2}{a} \arccos\left(-\frac{E + E_0}{2\gamma}\right).$$

Since  $E = \hbar^2 k^2 / 2m$ , so  $k = \sqrt{2mE/\hbar^2}$ ,  $\nabla_k E = \hbar^2 k / m$ ,  $\nabla_k E = \gamma a \sin(ka/2)$  for isotropic dispersion.

- (a) In a 3D situation, the general formula becomes

$$g(E) = \frac{2V}{(2\pi)^3} \cdot \frac{4\pi k^2}{\hbar^2 k / m} = \frac{Vm k}{\pi^2 \hbar^2} = \frac{Vm}{\pi^2 \hbar^3} \sqrt{2mE}.$$

For isotropic dispersion, the general formula becomes

$$g(E) = \frac{Vk^2}{\pi^2 \gamma a \sqrt{1 - \left(\frac{E+E_0}{2\gamma}\right)^2}}.$$

- (b) In a 2D situation, the general formula becomes

$$g(E) = \frac{2A}{(2\pi)^2} \cdot \frac{2\pi k}{\hbar^2 k / m} = \frac{mA}{\pi \hbar^2} = \text{Const.}$$

For isotropic dispersion, the general formula becomes

$$g(E) = \frac{2A}{(2\pi)^2} \cdot \frac{2\pi k}{\gamma a \sin(ka/2)} = \frac{2A}{\pi \gamma a^2} \frac{\arccos\left(-\frac{E+E_0}{2\gamma}\right)}{\sqrt{1 - \left(\frac{E+E_0}{2\gamma}\right)^2}}$$

- (c) In 1D situation,  $g(E) = \frac{2L}{2\pi} \cdot \frac{2}{dE/dk}$ . The general formula becomes

$$g(E) = \frac{2L}{2\pi} \cdot \frac{2}{dE/dk} = \frac{2L}{\pi dE/dk} = \frac{L}{\pi \hbar} \sqrt{\frac{2m}{E}}$$

For isotropic dispersion, the general formula becomes

$$g(E) = \frac{2L}{\pi \gamma a} \frac{1}{\sqrt{1 - \left(\frac{E+E_0}{2\gamma}\right)^2}}.$$



## Lecture #2 Phonon

**Problem 2.1** (Normal modes of a square lattice). Consider a square lattice of atoms of mass  $M$ , in which the nearest neighbors are coupled to each other by springs with constants  $K$ . Derive the equation of motion for the mode in which atomic displacements are normal to the crystal plane and find the dispersion of this mode. Analyze the behavior of this mode in the continuum limit.

**Solution.** Assume the lattice constant  $a$ , then the lattice sites become

$$\mathbf{R}_{n,m} = (na, ma), \quad n, m \in \mathbb{Z}.$$

We consider every atom has a single degree of freedom: the out-of-plane scalar displacement  $u_{n,m}(t)$ . So, the spring connecting site  $(n, m)$  and a neighbour  $(n', m')$  has extension  $\Delta = u_{n',m'} - u_{n,m}$ , which brings the force to atom  $(n, m)$

$$F_{n,m} = -K(u_{n,m} - u_{n',m'})$$

due to Hooke's law. Since every atom has four nearest neighbours:  $(n \pm 1, m)$ ,  $(n, m \pm 1)$ , which bring four forces to the “central atom”. Summing these forces

$$F_{n,m}^{\text{tot}} = - \sum_{\langle n,m;n',m' \rangle} K(u_{n,m} - u_{n',m'}) = -K(4u_{n,m} - u_{n+1,m} - u_{n-1,m} - u_{n,m+1} - u_{n,m-1})$$

According to Newton's second law, we obtain the EOM

$$M\ddot{u}_{n,m} = -K(4u_{n,m} - u_{n+1,m} - u_{n-1,m} - u_{n,m+1} - u_{n,m-1}).$$

Assume the EOM has a plane-wave solution

$$u_{n,m}(t) = A e^{i(k_x na + k_y ma - \omega t)},$$

then substitute it into the EOM, the LHS is

$$\text{LHS} = M(-\omega^2)u_{n,m}$$

and the RHS is

$$\text{RHS} = -K(4 - e^{ik_x a} - e^{-ik_x a} - e^{ik_y a} - e^{-ik_y a})u_{n,m} = -2K[2 - \cos(k_x a) - \cos(k_y a)]u_{n,m}$$

Finally we have the dispersion of this model

$$\omega^2(\mathbf{k}) = \frac{2K}{M}[2 - \cos(k_x a) - \cos(k_y a)], \quad \omega(\mathbf{k}) = 2\sqrt{\frac{K}{M}} \sqrt{\sin^2\left(\frac{k_x a}{2}\right) + \sin^2\left(\frac{k_y a}{2}\right)}.$$

In the continuum limit, i.e.,  $|\mathbf{k}|a \ll 1$ , then the dispersion of this model becomes linear to  $|\mathbf{k}|$

$$\omega(\mathbf{k}) \approx 2\sqrt{\frac{K}{M}} \sqrt{\left(\frac{k_x a}{2}\right)^2 + \left(\frac{k_y a}{2}\right)^2} = c|\mathbf{k}|$$

where  $c = a\sqrt{K/M}$ .

**Problem 2.2** (Localized phonons on an impurity). Consider a linear chain of identical atoms of mass  $m$  each equidistant by  $a$ .

- Limiting the problem to interactions between nearest neighbors characterized by the force constant  $\beta$ , give the dispersion relation of longitudinal phonons and derive the expression of their group velocity. Express the results as a function of “ $a$ ” and of the velocity of sound,  $v_s$ . What is maximum frequency of atomic vibration,  $\omega_M$ , when  $v_s = 5000 \text{ m/s}$ ?
- An atom in the middle of the chain is replaced by the atom of an impurity with a mass  $m'$  (where  $m' < m$ ). Write the displacement equations,  $u_0$ , for this impurity atom and the displacement “ $u_n$ ” for an atom located at a distance “ $na$ ”. Assume that the force constant  $\beta$  between nearest neighbors is the same regardless of the atom considered.
- The impurity atom being lighter than the others has an oscillation frequency  $\omega'_m$  larger than  $\omega_M$  and it drags the other neighbor atoms by its motion. We try to describe its effect with the expression of a damped wave of form:  $u_n = u_0 e^{-\alpha|x|} e^{i(\omega't - kx)}$ . What is the value of the frequency  $\omega'_M$ , which is compatible with the equations of motion when the wave vector is near the boundary of the first Brillouin Zone? Verify that when  $m = m'$ , the solution matches that obtained in (a).
- After having numerically evaluated  $\omega'_M$  and a when  $m' = m/2$ , sketch the instantaneous position of atoms as a function of  $x$ .

### Solution.

- Assume the nearest-neighbor spring constant  $\beta$ . The EOM for the  $n$ -th atom is

$$m\ddot{u}_n = -\beta(u_n - u_{n+1}) - \beta(u_n - u_{n-1}) = \beta(u_{n+1} - u_{n-1} - 2u_n).$$

We assume the EOM has a plane-wave solution  $u_n(t) = A e^{i(kna - \omega t)}$ , then substitute it into the EOM

$$-m\omega^2 u_n = -\beta(2 - e^{ika} - e^{-ika})u_n = -\beta(2 - 2\cos(ka))u_n = -4\beta \sin^2\left(\frac{ka}{2}\right),$$

we have the dispersion

$$\omega = 2\sqrt{\frac{\beta}{m}} \left| \sin\left(\frac{ka}{2}\right) \right|.$$

The group velocity and the velocity of sound are

$$v_g(k) = \frac{d\omega}{dk} = a\sqrt{\frac{\beta}{m}} \cos\left(\frac{ka}{2}\right), \quad \text{and} \quad v_s(a) = \lim_{k \rightarrow 0} v_g(k) = a\sqrt{\frac{\beta}{m}}$$

then the group velocity, as well as the frequency can be expressed as

$$v_g(k) = v_s \cos\left(\frac{ka}{2}\right), \quad \text{and} \quad \omega = \frac{2v_s}{a} \left| \sin\left(\frac{ka}{2}\right) \right|.$$

when  $k = \pi/a$ , the frequency arrives at a maximal value

$$\omega_M = \frac{2v_s}{a} \frac{v_s=5000 \text{ m/s}}{10000} \text{ s}^{-1}.$$

- (b) The impurity is located at site  $n = 0$ , with mass  $m'$ ; while its neighbors  $n = \pm 1$  with mass  $m$ . The EOM for the impurity and other atoms are

$$\begin{aligned} m' \ddot{u}_0 &= -\beta(u_1 - u_0) + \beta(u_{-1} - u_0) = \beta(u_1 + u_{-1} - 2u_0) \\ m \ddot{u}_n &= \beta(u_{n+1} + u_{n-1} - 2u_n) \end{aligned}$$

- (c) In (a), near the boundary of the first BZ, i.e.  $k = \pi/a$ , the frequency is  $\omega_1 = 2\sqrt{\beta/m}$ . Write the given wave form in terms of site number  $n$

$$u_n = A e^{-\alpha|n|a} e^{i(\omega't - kna)}$$

Substitute it into the two EOMs in (b), and let  $n = 1$

$$\begin{aligned} -\omega^2 m' u_0 &= \beta(e^{-\alpha a - ika} + e^{-\alpha a + ika} - 2)u_0 \stackrel{k=\pi/a}{=} -2\beta(e^{-\alpha a} + 1)u_0 \\ -\omega^2 m u_1 &= \beta(u_2 + u_0 - 2u_1) = \beta(e^{-\alpha a - ika} + e^{\alpha a + ika} - 2)u_1 \stackrel{k=\pi/a}{=} -2\beta(\cosh(\alpha a) + 1)u_1 \end{aligned}$$

Derive the two expressions, we have

$$\frac{m'}{m} = \frac{e^{-\alpha a} + 1}{\cosh(\alpha a) + 1} = \frac{2}{e^{\alpha a} + 1}, \quad \text{and} \quad e^{-\alpha a} + 1 = \frac{2m}{2m - m'}$$

Substitute it into one of the EOM, we obtain the dispersion

$$\omega_M'^2 = \omega'^2 = \frac{4\beta m}{m'(2m - m')}$$

where  $\omega_M'^2 = \omega'^2$  is due to the boundary of the first BZ ( $k = \pi/a$ ). When  $m' = m$ , the frequency becomes

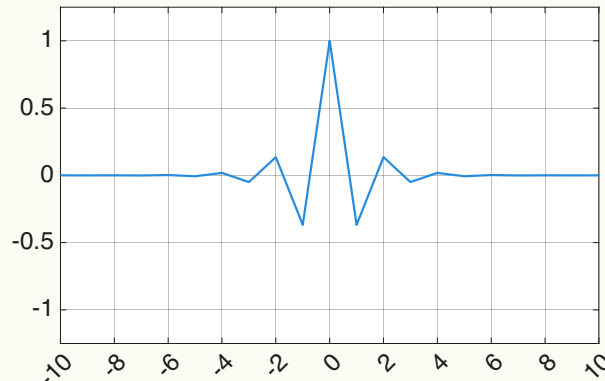
$$\omega_M'^2 = \frac{4\beta}{m}, \quad \text{and} \quad \omega_M' = 2\sqrt{\frac{\beta}{m}} = \omega_1$$

which matches that obtained in (a).

- (d) Substitute  $m' = m/2$  in the result of (c), we have  $\omega_M'^2 = \frac{16}{3} \frac{\beta}{m}$ . To simplify, we choose the time  $t = 0$ . Near the first BZ, the displacement of atoms is

$$u_n = (-1)^n A e^{-\alpha|n|a}$$

The plotted figure of atoms is attached as follows (Just show the relative position for reference).



**Problem 2.3** (Density of states and specific heat of a monoatomic 1D lattice). Consider a row of  $N$  identical atoms of mass  $m$  and equidistant by  $a$ .

- Limiting the problem to interactions between nearest neighbors with a force constant  $\beta$ , find the expression for the dispersion relation of acoustic longitudinal vibrations propagating along this row. What is the maximum frequency  $\omega_M$  of atomic vibrations, written in terms of  $v_s$  and  $a$ ? Compare the value obtained with that of  $\omega_D$  deduced from the Debye model.
- Find the corresponding expression for the density of states  $g(\omega)$ .
- Write the expressions for internal energy  $U$  and for the heat capacity  $C_v$ . Find the limits as these functions approach high temperatures ( $\hbar\omega_M \ll k_B T$ ).
- What is the evolution of  $C_v$  at low temperature? Compare the result obtained here with that deduced from the Debye model.

**Solution.**

- The same to Problem 2.2(a): the dispersion

$$\omega(k) = 2\sqrt{\frac{\beta}{m}} \left| \sin\left(\frac{ka}{2}\right) \right|, \quad v_s = \lim_{k \rightarrow 0} \frac{d\omega}{dk} = a\sqrt{\frac{\beta}{m}}, \quad \text{and} \quad \omega_M \stackrel{k=\pi/a}{=} 2\sqrt{\frac{\beta}{m}} = \frac{2v_s}{a},$$

and the Debye frequency for 1D chain is  $\omega_D = v_s k = \frac{\pi v_s}{a}$ . Since  $\omega_M/\omega_D = 2/\pi < 1$ , the maximum frequency in the real dispersion is smaller than in the Debye model.

- In 1D chain, the number of states within  $dk$  is  $2 \cdot \frac{L}{2\pi} dk$ . So, the density of states is

$$g(\omega) = \frac{L}{\pi} \frac{dk}{d\omega} = \frac{2L}{\pi a \omega_M} \frac{1}{\sqrt{1 - (\omega/\omega_M)^2}} \stackrel{L=Na}{=} \frac{2N}{\pi \omega_M} \frac{1}{\sqrt{1 - (\omega/\omega_M)^2}}.$$

- The phonon thermal occupation

$$n(\omega) = \left[ \exp\left(\frac{\hbar\omega}{k_B T}\right) - 1 \right]^{-1}$$

Then the internal energy is

$$U(T) = \int_0^{\omega_M} \hbar\omega n(\omega) g(\omega) d\omega = \int_0^{\omega_M} \frac{\hbar\omega}{e^{\frac{\hbar\omega}{k_B T}} - 1} g(\omega) d\omega.$$

and the heat capacity

$$C_V(T) = \frac{dU}{dT} = \int_0^{\omega_M} \frac{\partial n(\omega)}{\partial T} g(\omega) d\omega$$

At high-temperature limit, i.e.,  $\hbar\omega_M \ll k_B T$ . The expression of internal energy and heat capacity become

$$U(T) = \int_0^{\omega_M} k_B T g(\omega) d\omega = N k_B T, \quad C_V(T) = N k_B.$$

where applied the total number of vibration modes  $\int_0^{\omega_M} g(\omega) d\omega = N$ .

- (d) At low temperature limit, only small  $k$  matters, i.e., the main contributions come from small  $\omega$ . Then,  $g(\omega) = \frac{L}{\pi v_s}$ . The internal energy becomes

$$U = \frac{L}{\pi v_s} \int_0^\infty \frac{\hbar\omega}{e^{\frac{\hbar\omega}{k_B T}} - 1} d\omega = \frac{\pi L (k_B T)^2}{6\hbar v_s}$$

and the capacity

$$C_v = \frac{dU}{dT} = \frac{\pi N a k_B^2 T}{3\hbar v_s} \propto T$$

the same conclusion as the Debye model.

## Lecture #3 Crystal lattice

**Problem 3.1** (Crystallographic restriction theorem). Proof the crystallographic restriction theorem: Consider a crystal (i.e. with translational symmetry) in three dimensions that is invariant under rotations by an angle  $2\pi/n$  around an axis. Then the only allowed possibilities are  $n = 1, 2, 3, 4, 6$  with the case  $n = 1$  being the trivial case with no rotational symmetry.

**Solution.** *Proof.* Denote the 3D lattice can be generated by the basis vectors  $e_1, e_2$ , and  $e_3$ . So, any lattice point vector set should be

$$\mathbf{L} = \{a_1\mathbf{e}_1 + a_2\mathbf{e}_2 + a_3\mathbf{e}_3\}.$$

Obviously,  $a_i \in \mathbb{Z}$ , i.e., any lattice point can be labeled by integer coordinates  $(a_1, a_2, a_3)$ . Consider the 3D rotation matrix (operator)

$$\mathbf{R} = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

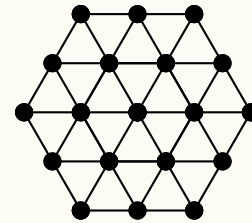
When acting it on the vector  $\mathbf{L}$ , it should leave the lattice invariant, i.e., the trace must be integers

$$\text{Tr}(\mathbf{R}) = 2 \cos \theta + 1 \in \mathbb{Z}.$$

So we have  $2 \cos \theta = 2 \cos(2\pi/n) \in \mathbb{Z}$ , i.e.,  $\cos(2\pi/n) = -1, -1/2, 0, 1/2, 1$ , then  $n \in \{1, 2, 3, 4, 6\}$ .  $\square$

**Problem 3.2** (2D triangular lattice). Let us consider a two-dimensional triangular lattice as shown in following Figure.

- Identify the primitive unit cell of the two-dimensional triangular lattice. Find the basis vectors.
- Construct the basis vectors of the reciprocal unit cell.



**Solution.**

- Denote the lattice constant  $a$ . Then the simplest choice of primitive lattice vectors is

$$\mathbf{a}_1 = a(1, 0), \quad \mathbf{a}_2 = a\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right).$$

- Denote the two reciprocal-lattice primitive vectors  $\mathbf{b}_1 = (b_{1x}, b_{1y})$ ,  $\mathbf{b}_2 = (b_{2x}, b_{2y})$ . Using the relation

$$\mathbf{a}_i \cdot \mathbf{b}_j = 2\pi\delta_{ij}.$$

To expand it

$$\mathbf{a}_1 \cdot \mathbf{b}_1 = 2\pi, \quad \mathbf{a}_2 \cdot \mathbf{b}_1 = 0, \quad \mathbf{a}_1 \cdot \mathbf{b}_2 = 0, \quad \mathbf{a}_2 \cdot \mathbf{b}_2 = 2\pi,$$

we obtain the two reciprocal-lattice primitive vectors

$$\mathbf{b}_1 = \frac{2\pi}{a}\left(1, -\frac{1}{\sqrt{3}}\right), \quad \mathbf{b}_2 = \frac{2\pi}{a}\left(0, \frac{2}{\sqrt{3}}\right).$$

**Problem 3.3** (Properties of the reciprocal lattice).

- (a) Show that all reciprocal lattice vectors of the form  $\mathbf{G} = h\mathbf{A} + k\mathbf{B} + l\mathbf{C}$  are perpendicular to the planes of the same indices  $(h, k, l)$  in real space.
- (b) Show that the distance  $d_{h,k,l}$  between two consecutive planes  $(h, k, l)$  is inversely proportional to  $|\mathbf{G}_{h,k,l}|$ .
- (c) Find  $d_{h,k,l}$  for a simple cubic lattice and an orthorhombic lattice ( $a \neq b \neq c, \alpha = \beta = \gamma = \pi/2$ ).

**Solution.** “

- (a) *Proof.* Substitute the expressions of the reciprocal lattice vectors to  $\mathbf{G}$

$$\mathbf{G} = \frac{2\pi}{V} [h(\mathbf{a}_2 \times \mathbf{a}_3) + k(\mathbf{a}_3 \times \mathbf{a}_1) + l(\mathbf{a}_1 \times \mathbf{a}_2)].$$

Since the Miller indices  $(h, k, l)$  correspond to a plane that intercepts the crystal axes at  $\mathbf{a}_1/h$ ,  $\mathbf{a}_2/k$ ,  $\mathbf{a}_3/l$ , then, any vector lying in the  $(h, k, l)$  plane can be written as a linear combination of two independent vectors in that plane, i.e.,

$$\mathbf{r}_1 = \frac{\mathbf{a}_2}{k} - \frac{\mathbf{a}_1}{h}, \quad \mathbf{r}_2 = \frac{\mathbf{a}_3}{l} - \frac{\mathbf{a}_2}{k}.$$

Then, the normal to this plane can be written as a cross product

$$\hat{\mathbf{n}}_{hkl} = \mathbf{r}_1 \times \mathbf{r}_2 = \frac{1}{kl}(\mathbf{a}_2 \times \mathbf{a}_3) + \frac{1}{hl}(\mathbf{a}_3 \times \mathbf{a}_1) + \frac{1}{hk}(\mathbf{a}_1 \times \mathbf{a}_2).$$

which is parallel to  $\mathbf{G}$ , implies that  $\mathbf{G}$  is perpendicular to the  $(h, k, l)$  plane.  $\square$

- (b) *Proof.* For the vector pointing to the  $n+1$  and  $n$ -th  $(h, k, l)$  plane, it satisfies

$$\mathbf{G} \cdot \mathbf{r}_{n+1} = 2\pi(n+1), \quad \text{and} \quad \mathbf{G} \cdot \mathbf{r}_n = 2\pi n,$$

where  $\mathbf{r}_{n+1} - \mathbf{r}_n = d\hat{\mathbf{n}}$ . Combine the two relations

$$\mathbf{G} \cdot (\mathbf{r}_{n+1} - \mathbf{r}_n) = \mathbf{G} \cdot \hat{\mathbf{n}}d = 2\pi d.$$

So, we have

$$d = \frac{2\pi}{\mathbf{G} \cdot \hat{\mathbf{n}}} = \frac{2\pi}{|\mathbf{G}_{h,k,l}|},$$

i.e., the distance between two consecutive planes is inversely proportional to  $|\mathbf{G}_{h,k,l}|$ .  $\square$

- (c) For a simple cubic lattice

$$\mathbf{a} = (a, 0, 0), \quad \mathbf{b} = (0, b, 0), \quad \mathbf{c} = (0, 0, c).$$

Then, the reciprocal vectors can be expressed as

$$\mathbf{G} = \frac{2\pi h}{a}\hat{\mathbf{x}} + \frac{2\pi k}{b}\hat{\mathbf{y}} + \frac{2\pi l}{c}\hat{\mathbf{z}},$$

and its magnitude

$$|\mathbf{G}| = 2\pi \sqrt{\frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}}.$$

So, we obtain the distance  $d = \frac{2\pi}{|\mathbf{G}|} = \left( \frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2} \right)^{-1/2}$ .

**Problem 3.4** (Atomic planes and Miller indices: application to lithium). The Bravais lattice of lithium is simple cubic with lattice parameter  $a = 3.48 \text{ \AA}$ .

- Suppose that the atoms (assumed to be spheres) are placed along the  $[111]$ , represent the atomic distribution along the following faces (100), (110), (111), and (201).
- For each such 2D structure, state the direction and basis vector,  $\mathbf{a}$  and  $\mathbf{b}$  of the elementary lattice as well as the value of the angle,  $\gamma$ .
- Find the atomic concentration and the mass density of lithium ( $A \approx 7$ ).

**Solution.**

- (b) The distribution, basis vector, and the angle of the faces are listed.

| 2D Structure | Distribution | $\mathbf{a}$ | $\mathbf{b}$  | $\gamma$   |
|--------------|--------------|--------------|---------------|------------|
| (100)        | square       | $(0, a, 0)$  | $(0, 0, a)$   | $90^\circ$ |
| (110)        | rectangular  | $(a, -a, 0)$ | $(0, 0, a)$   | $90^\circ$ |
| (111)        | hexagonal    | $(a, 0, -a)$ | $(0, a, -a)$  | $60^\circ$ |
| (201)        | rectangular  | $(0, a, 0)$  | $(a, 0, -2a)$ | $90^\circ$ |

- Since the atoms are placed along the  $[111]$ , then every cubic lattice's corners has an atom, i.e., it is equivalent that there is an atom in every lattice. So, the atomic concentration is

$$n = \frac{1}{V} = \frac{1}{a^3} = 2.3728 \times 10^{28} \text{ atoms/m}^3.$$

The mass density is

$$\rho = nm_{\text{atom}} = \frac{nA}{N_A} = 275.816 \text{ kg/m}^3.$$



## Lecture #4 X-ray diffraction

**Problem 4.1** (Structure factor for a tri-atomic basis). Consider a linear crystal with lattice constant  $a$  in which the basis consists of atom of species  $A$  at 0 and two atoms of species  $B$  at  $1/3$  and  $2/3$ .

- Find the structure factor of such a crystal.
- X-rays of wavelength  $\lambda (= a/4)$  irradiate this crystal at an oblique incidence ( $45^\circ$ ). Using the Ewald construction in the plane of incidence, show graphically, the diffraction directions and indicate the relative weights of the different diffracted intensities.
- State the physical significance of the vector  $\mathbf{k}$  from which the extremity is on a plane passing through the origin of the reciprocal lattice.

**Solution.**

- Substitute  $r_i = 0, \frac{a}{3}, \frac{2a}{3}$  into the definition of the structure factor

$$S(h) = f_A e^{iG \cdot 0} + f_B e^{iG \cdot \frac{a}{3}} + f_B e^{iG \cdot \frac{2a}{3}} \xrightarrow{G=2\pi h/a} f_A + f_B \left( e^{i\frac{2\pi h}{3}} + e^{i\frac{4\pi h}{3}} \right)$$

$$\xrightarrow{\text{Euler's formula}} \begin{cases} f_A + 2f_B, & h = 3n \\ f_A - f_B, & \text{otherwise} \end{cases}$$

- The wave number

$$k = \frac{2\pi}{\lambda} = \frac{8\pi}{a} = 4b_0,$$

where  $b_0 = \frac{2\pi}{a}$  is the reciprocal lattice vector. The incident wavevector is

$$\mathbf{k}_i = \left( k/\sqrt{2}, -k/\sqrt{2} \right)$$

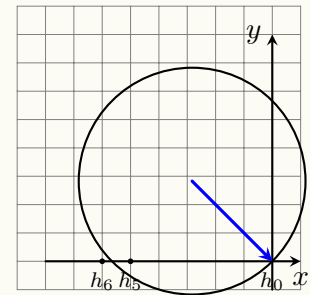
Then, the Ewald circle satisfies

$$\left( x + k/\sqrt{2} \right)^2 + \left( y - k/\sqrt{2} \right)^2 = k^2$$

and we can obtain the two intersection on  $x$ -axis:  $x_1 = 0$  and

$$x_1 = -\sqrt{2}k.$$

Obviously,  $x_1 = 0b_0, n_1 = 0; x_2 = -5.656b_0, n_2 = -5.656$ .



Since  $n$  is not an integer, no other reciprocal lattice points lies on the Ewald circle except the one on the original point. The intensity of each diffraction order is proportional to  $|S(h)|^2$ , i.e.,

$$I_h \propto |S(h)|^2 = \begin{cases} (f_A + 2f_B)^2, & h = 3m, \\ (f_A - f_B)^2, & \text{otherwise} \end{cases}$$

- The tip of  $\mathbf{k}$  lying on a plane through the origin of the reciprocal lattice means the Bragg condition is

satisfied for diffraction.

**Problem 4.2.** The main experimental methods of X-ray diffraction are

- (a) the Laue method
- (b) the crystal rotation method
- (c) the powder method (or the Debye-Scherrer method)

Describe the corresponding experimental set-up and with the help of the Ewald construction in reciprocal space, deduce the observed diffraction patterns in each case.

**Solution.**

(a) Laue Method

- i. Setup: Stationary single crystal, polychromatic X-rays.
- ii. Ewald & Pattern: A continuous range of Ewald sphere radii (due to varying  $\lambda$ ) means many reciprocal lattice points lie on some sphere. This produces a pattern of sharp spots, each from a different  $(hkl)$  plane and wavelength. The spot arrangement directly reveals the crystal's symmetry.

(b) Crystal Rotation Method

- i. Setup: Rotating single crystal, monochromatic X-rays.
- ii. Ewald & Pattern: The fixed Ewald sphere is intersected by rotating reciprocal lattice points. This generates discrete spots or arcs on the detector. Measuring the rotation angles and intensities of these spots allows for the determination of the unit cell and atomic structure.

(c) Powder Method

- i. Setup: Powdered sample (random crystallites), monochromatic X-rays.
- ii. Ewald & Pattern: Each set of  $(hkl)$  planes forms a spherical shell in reciprocal space. The Ewald sphere's intersection with these shells creates cones of diffracted beams (Debye cones), which appear as characteristic rings on the detector. The ring positions ( $2\theta$  angles) provide  $d$ -spacings for phase identification and lattice parameter calculation.

## Lecture #5 Nearly free electron model

**Problem 5.1** (Compare Kronig-Penney model with the nearly free electron model). The Kronig-Penney model allows one to consider both the nearly-free-electron and tight-binding limits. Consider the (repulsive) Dirac-Kronig-Penney model. It gives the following implicit equation for the energy bands

$$\cos(ka) = \cos(qa) + u \frac{\sin(qa)}{qa}$$

where  $q = \sqrt{2mE}/\hbar$  and  $u = \frac{mU_0a}{\hbar^2} > 0$ . Assuming that  $u \ll q$ , find the solution of above equation near the Bragg points  $k \approx \pm \frac{\pi}{a}, \pm \frac{2\pi}{a}, \dots$ . Compare your result with that of the nearly-free electron model. (Reminder: the  $\approx$  sign means that  $k$  is close yet not equal to the wavenumber of the Bragg point.)

**Solution.** Near the Bragg points, assume

$$ka = n\pi + \kappa a, \quad \text{and} \quad qa = q_n a + \delta.$$

with  $|\kappa|, |\delta| \ll 1$ ,  $q_n = n\pi/a$ . Keep the 2nd order of  $\kappa$  and  $\delta$ , the implicit equation can be written as

$$(-1)^n \left[ 1 - \frac{1}{2}(\kappa a)^2 \right] = (-1)^n \left( 1 - \frac{1}{2}\delta^2 \right) + u \frac{(-1)^n \delta}{n\pi}.$$

Then, we can have the solution of  $\delta$

$$\delta = \frac{u}{n\pi} \pm \sqrt{\left( \frac{u}{n\pi} \right)^2 + (\kappa a)^2}.$$

Since  $q = \sqrt{2mE}/\hbar$ , then we have

$$E = \frac{\hbar^2(q_n + \delta/a)^2}{2m} \approx E_n^0 + \frac{\hbar^2 q_n \delta/a}{m},$$

where  $E_n^0 = \frac{\hbar^2 q_n^2}{2m}$ . Substitute  $q_n = n\pi/a$ , and the solution of  $\delta$  into the expression of  $E$

$$E = E_n^0 + \frac{n\pi\hbar^2\delta}{ma} = E_n^0 + \frac{n\pi\hbar^2}{ma^2} \left[ \frac{u}{n\pi} \pm \sqrt{\left( \frac{u}{n\pi} \right)^2 + (\kappa a)^2} \right].$$

Substitute  $u = mU_0a/\hbar^2$  into the expression above

$$E = E_n^0 + \frac{U_0}{a} \pm \sqrt{\left( \frac{U_0}{a} \right)^2 + \left( \frac{n\pi\hbar^2\kappa}{ma} \right)^2},$$

with the band gap

$$\Delta E_n = E_+ - E_- = 2\sqrt{\left( \frac{U_0}{a} \right)^2 + \left( \frac{n\pi\hbar^2\kappa}{ma} \right)^2}.$$

When  $\kappa a \rightarrow 0$ , the energy band and the band gap become

$$E = E_n^0 + \frac{U_0}{a} \pm \frac{U_0}{a}, \quad \text{and} \quad \Delta E_n = \frac{2U_0}{a}.$$

The results perfectly match that of the nearly-free electron model.

**Problem 5.2** (Nearly free electron model). Consider 2D electrons subject to a weak periodic potential

$$U(x, y) = U_0 \left( \cos \frac{2\pi x}{a} + \cos \frac{2\pi y}{a} \right).$$

- (a) Find the energy bands near the edges of the first Brillouin zone. (Hint: the edges of the Brillouin zone are the Bragg planes where the eigenstates are doubly degenerate.) Plot the bands and isoenergetic surfaces.
- (b) Repeat the analysis of part (a) for regions near the corners of the first Brillouin zone, where there are four degenerate eigenstates.

**Solution.**

- (a) In reciprocal space, the reciprocal lattice vectors are

$$\mathbf{G}_1 = \left( \frac{2\pi}{a}, 0 \right), \quad \mathbf{G}_2 = \left( 0, \frac{2\pi}{a} \right).$$

The potential can be expressed as

$$U(x, y) = \frac{U_0}{2} \left[ e^{i\mathbf{G}_1 \cdot \mathbf{r}} + e^{-i\mathbf{G}_1 \cdot \mathbf{r}} + e^{i\mathbf{G}_2 \cdot \mathbf{r}} + e^{-i\mathbf{G}_2 \cdot \mathbf{r}} \right].$$

At the zone edge (e.g.,  $k_x = \pi/a$ ), consider a wavevector

$$\mathbf{k} = \left( \frac{\pi}{a} + q_x, k_y \right),$$

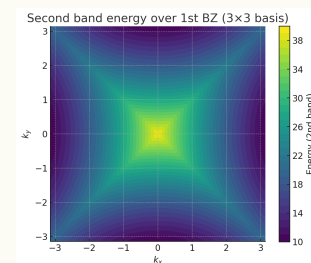
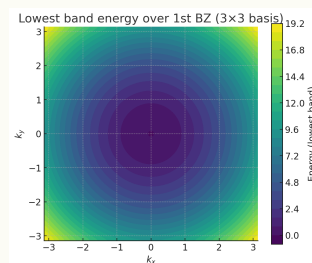
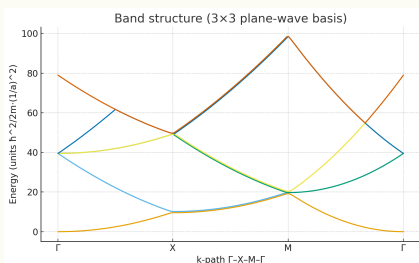
with  $|q_x| \ll \pi/a$ . The two nearly degenerate plane waves are  $|\mathbf{k}\rangle$  and  $|\mathbf{k} - \mathbf{G}_1\rangle$ . Their free-electron energies are

$$E^{(0)}(\mathbf{k}) = \frac{\hbar^2}{2m} \left[ \left( \frac{\pi}{a} + q_x \right)^2 + k_y^2 \right],$$

$$E^{(0)}(\mathbf{k} - \mathbf{G}_1) = \frac{\hbar^2}{2m} \left[ \left( -\frac{\pi}{a} + q_x \right)^2 + k_y^2 \right].$$

Expanding to first order in  $q_x$

$$E^{(0)} \approx E_0 \pm \alpha q_x, \quad \text{where} \quad E_0 = \frac{\hbar^2}{2m} \left[ \left( \frac{\pi}{a} \right)^2 + k_y^2 \right], \quad \alpha = \frac{\hbar^2 \pi}{ma}.$$



The perturbation couples these states with matrix element  $\langle \mathbf{k} | U | \mathbf{k} - \mathbf{G}_1 \rangle = U_0/2$ . The Hamiltonian in the subspace  $\{|\mathbf{k}\rangle, |\mathbf{k} - \mathbf{G}_1\rangle\}$  is

$$H = \begin{pmatrix} E_0 + \alpha q_x & U_0/2 \\ U_0/2 & E_0 - \alpha q_x \end{pmatrix}.$$

Diagonalizing gives the eigenenergies

$$E_{\pm} = E_0 \pm \sqrt{(\alpha q_x)^2 + \left(\frac{U_0}{2}\right)^2} = E_0 + \frac{\hbar^2 q_x^2}{2m} \pm \sqrt{\left(\frac{\hbar^2 \pi}{ma} q_x\right)^2 + \left(\frac{U_0}{2}\right)^2}$$

- (b) The corners of the first Brillouin zone are points like  $\mathbf{k}_0 = (\pi/a, \pi/a)$  (the M point). The four degenerate plane waves at this point are

$$\begin{aligned} \mathbf{k}_1 &= \mathbf{k}_0 = (+\pi/a, +\pi/a), \\ \mathbf{k}_2 &= \mathbf{k}_0 - \mathbf{G}_1 = (-\pi/a, +\pi/a), \\ \mathbf{k}_3 &= \mathbf{k}_0 - \mathbf{G}_2 = (+\pi/a, -\pi/a), \\ \mathbf{k}_4 &= \mathbf{k}_0 - \mathbf{G}_1 - \mathbf{G}_2 = (-\pi/a, -\pi/a). \end{aligned}$$

All have the same free-electron energy  $E_0 = \frac{\hbar^2}{2m} [(\pi/a)^2 + (\pi/a)^2] = \frac{\hbar^2 \pi^2}{ma^2}$ .

In the basis  $\{|\mathbf{k}_1\rangle, |\mathbf{k}_2\rangle, |\mathbf{k}_3\rangle, |\mathbf{k}_4\rangle\}$ , states are coupled if they differ by  $\pm \mathbf{G}_1$  or  $\pm \mathbf{G}_2$ . The non-zero off-diagonal matrix elements are

$$\langle \mathbf{k}_1 | U | \mathbf{k}_2 \rangle = \langle \mathbf{k}_1 | U | \mathbf{k}_3 \rangle = \langle \mathbf{k}_2 | U | \mathbf{k}_4 \rangle = \langle \mathbf{k}_3 | U | \mathbf{k}_4 \rangle = U_0/2.$$

The Hamiltonian is

$$H = E_0 \mathbb{1} + \frac{U_0}{2} \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}.$$

Diagonalizing this matrix yields eigenvalues

$$E = E_0 + \frac{U_0}{2} \times \{2, -2, 0, 0\} = \{E_0 + U_0, E_0 - U_0, E_0, E_0\}.$$

Thus, at  $\mathbf{q} = 0$ , the fourfold degeneracy splits into three distinct levels: one non-degenerate band at  $E_0 + U_0$ , one non-degenerate band at  $E_0 - U_0$ , and a two-fold degenerate band at  $E_0$ .

## Lecture #6 Tight binding model

**Problem 6.1** (Tight bindings in the BCC and FCC lattices). We consider identical atoms located at points of a lattice that are body-centered cubic (**BCC: i**) and face-centered cubic (**FCC: ii**).

- (a) For each case, determine the dispersion relation for  $s$ -electrons, having the general form

$$E = -\alpha - \gamma \sum_s e^{-i\mathbf{k} \cdot \mathbf{R}_s},$$

where  $\alpha$  and  $\gamma$  are the positive energies that can be calculated and  $\mathbf{R}_s$  represents the vectors that connect an atom located at the origin to each of its nearest neighbors.

- (b) Find the expressions for  $E$  at the points located at the intersection of the first Brillouin zone with the directions  $[100]$ ,  $[110]$ , and  $[111]$  and thus the value  $E(\Gamma)$  to the center of this zone.
- (c) Deduce the total width of the corresponding band,  $\Delta E$ , and find the expression for the effective mass  $m^*$  in the vicinity of  $\Gamma$  as well as the shape of constant energy surfaces around this point.

**Solution** (i = BCC, ii = FCC). Assume the lattice constant  $a$ .

- (a<sub>i</sub>) Each atom has 8 NNs, i.e.,

$$\{\mathbf{R}_s\} = \left\{ \frac{a}{2}(\pm 1, \pm 1, \pm 1) \right\}$$

Then, the sum in the general form becomes

$$\sum_s e^{-i\mathbf{k} \cdot \mathbf{R}_s} = \sum_{\sigma_x=\pm} \sum_{\sigma_y=\pm} \sum_{\sigma_z=\pm} e^{-i\frac{a}{2}\sigma_x k_x} \cdot e^{-i\frac{a}{2}\sigma_y k_y} \cdot e^{-i\frac{a}{2}\sigma_z k_z} = 8 \cos\left(\frac{ak_x}{2}\right) \cos\left(\frac{ak_y}{2}\right) \cos\left(\frac{ak_z}{2}\right)$$

Hence, the dispersion relation is

$$E(\mathbf{k}) = -\alpha - 8\gamma \cos\left(\frac{ak_x}{2}\right) \cos\left(\frac{ak_y}{2}\right) \cos\left(\frac{ak_z}{2}\right)$$

- (b<sub>i</sub>) i.  $\Gamma$  point:  $\mathbf{k} = (0, 0, 0)$ ,  $E(\Gamma) = -\alpha - 8\gamma$ .      iii.  $[110]$ :  $\mathbf{k} = \frac{2\pi}{a}(\frac{1}{2}, \frac{1}{2}, 0)$ ,  $E(N) = -\alpha$ .  
 ii.  $[100]$ :  $\mathbf{k} = \frac{2\pi}{a}(1, 0, 0)$ ,  $E(X) = -\alpha + 8\gamma$ .      iv.  $[111]$ :  $\mathbf{k} = \frac{\pi}{a}(1, 1, 1)$ ,  $E(P) = -\alpha$

- (c<sub>i</sub>) The cos terms in the general form differs between  $[-8\gamma, +8\gamma]$ . So, the energy gap is

$$\Delta E_{\text{BCC}} = (-\alpha + 8\gamma) - (-\alpha - 8\gamma) = 16\gamma$$

The dispersion relation can be expanded around the  $\Gamma$  point

$$E(\mathbf{k}) = -\alpha - 8\gamma \left[ 1 - \frac{1}{2} \left( \frac{ak_x}{2} \right)^2 - \frac{1}{2} \left( \frac{ak_y}{2} \right)^2 - \frac{1}{2} \left( \frac{ak_z}{2} \right)^2 \right] \stackrel{k^2=k_x^2+k_y^2+k_z^2}{=} -\alpha - 8\gamma + \gamma a^2 k^2$$

Then, the effective mass  $m^* = \frac{\hbar^2}{\partial^2 E / \partial k^2} = \frac{\hbar^2}{2\gamma a^2}$ .

(a<sub>ii</sub>) Each atom has 12 NNs, i.e.,

$$\{\mathbf{R}_s\} = \left\{ \frac{a}{2}(\pm 1, \pm 1, 0), \frac{a}{2}(\pm 1, 0, \pm 1), \frac{a}{2}(0, \pm 1, \pm 1) \right\}$$

Then, the sum in the general form becomes

$$\begin{aligned} \sum_s e^{-i\mathbf{k} \cdot \mathbf{R}_s} &= \sum_{\text{cyclic}(x,y,z)} \sum_{\pm} e^{-i\frac{a}{2}(\pm k_i \pm k_j)} \\ &= 4 \left[ \cos\left(\frac{ak_x}{2}\right) \cos\left(\frac{ak_y}{2}\right) + \cos\left(\frac{ak_y}{2}\right) \cos\left(\frac{ak_z}{2}\right) + \cos\left(\frac{ak_z}{2}\right) \cos\left(\frac{ak_x}{2}\right) \right] \end{aligned}$$

Hence, the dispersion relation is

$$E(\mathbf{k}) = -\alpha - 4\gamma \left[ \cos\left(\frac{ak_x}{2}\right) \cos\left(\frac{ak_y}{2}\right) + \cos\left(\frac{ak_y}{2}\right) \cos\left(\frac{ak_z}{2}\right) + \cos\left(\frac{ak_z}{2}\right) \cos\left(\frac{ak_x}{2}\right) \right]$$

- (b<sub>ii</sub>) i.  $\Gamma$  point:  $\mathbf{k} = (0, 0, 0)$ ,  $E(\Gamma) = -\alpha - 8\gamma$ .    ii.  $[100]$ :  $\mathbf{k} = \frac{2\pi}{a}(1, 0, 0)$ ,  $E(X) = -\alpha - 4\gamma$ .  
 iii.  $[110]$  point:  $\mathbf{k} = \frac{2\pi}{a}(\frac{3}{4}, \frac{3}{4}, 0)$ ,  $E(N) = -\alpha + \gamma(4\sqrt{2} - 2)$ .  
 iv.  $[111]$  point:  $\mathbf{k} = \frac{\pi}{a}(1, 1, 1)$ ,  $E(P) = -\alpha - 12\gamma$

(c<sub>ii</sub>) The cos terms in the general form differs between  $[-12\gamma, +4\gamma]$ . So, the energy gap is

$$\Delta E_{\text{FCC}} = (-\alpha + 4\gamma) - (-\alpha - 12\gamma) = 16\gamma$$

The dispersion relation can be expanded around the  $\Gamma$  point

$$\begin{aligned} E(\mathbf{k}) &= -\alpha - 4\gamma \sum_{\text{cyclic}(x,y,z)} \left[ 1 - \frac{1}{2} \left( \frac{ak_i}{2} \right)^2 - \frac{1}{2} \left( \frac{ak_j}{2} \right)^2 \right] \\ &= -\alpha - 12\gamma + 4\gamma \left[ \left( \frac{k_x a}{2} \right)^2 + \left( \frac{k_y a}{2} \right)^2 + \left( \frac{k_z a}{2} \right)^2 \right] = -\alpha - 12\gamma + \gamma a^2 k^2 \end{aligned}$$

Then, the effective mass  $m^* = \frac{\hbar^2}{\partial^2 E / \partial k^2} = \frac{\hbar^2}{2\gamma a^2}$ .